## Fertilization, Implantation and Hormonal Control

- Analyse the features of fertilisation, implantation and hormonal control of pregnancy and birth in mammals.

**Important Terms:**

- Haploid → Containing one set of paired chromosomes.
- Diploid → Plant classification group where seeds are formed on flowers.
- Gametophyte → Plant classification group where seeds are formed on cones.
- Sporophyte → Cell division of germ cells that produces four non-identical haploid gametes.
- Meiosis → Cell division of Somatic cells that produces two identical new diploid cells.
- Mitosis → Gamete-forming haploid stage in a plant's life cycle.
- Angiosperm → Spore-forming diploid stage in a plant's life cycle.

**Fertilisation:**

- Mammals reproduce sexually using internal fertilisation.
- Through the fusion of male and female gametes (sperm and eggs) enabling the formation of a new organism.

Process –

- Ovary releases an egg (ovulation) about once a month in humans.
- The released egg travels to the fallopian tube via the oviduct.
- Sexual intercourse occurs and the male releases semen, which is comprised of sperm, directly into the vagina.
- The sperm swim from the vagina, through the cervix and uterus into the oviduct/fallopian tube.
- This is where the sperm fuses with the egg to create a zygote (new organism).
- Zygote creates a strong membrane which prevents any other sperm fertilising the egg meaning that the remaining sperm will die out.
- The zygote travels down the oviduct to the uterus where it is implanted.

**Implantation:**

- Implantation is the attachment of the fertilised egg to the lining of the uterus. (It buries itself deeper to be overcome by the lining).
- As an embryo form (looks human like), the amniotic sac, umbilical cord and placenta develop.
- Amniotic sac contains the unborn baby and is filled with fluid.
- keeps the baby in optimal temperature and provides safety from external factors.
- Placenta provides the oxygen and nutrients that a baby needs to survive whilst removing carbon dioxide and other waste. It is outside of the amniotic sac therefore there needs to be something to facilitate the nutrition/oxygen.
- The umbilical cord is the connection between the amniotic sac and the placenta (enables the survival of the baby).
- From implantation to birth, this period is known as pregnancy.
- Human pregnancy is about 9 months.

**Hormonal Control of Pregnancy and Birth:**

- Pregnancy and birth can only occur with the help of hormones (chemical messengers).

- Oestrogen → is caused by luteinising hormone (hormone which triggers ovulation) responsible for the development of female characteristics and stimulates the female body to release an egg. It also aids blood flow and enables the growth and development of vital organs (e.g., lungs and kidneys). It is later secreted in pregnancy to enable the production of progesterone.
- Progesterone → initially released by the ovaries and then later by the placenta. It stimulates the thickening of the uterine lining early in pregnancy. It gradually rises as pregnancy continues, this enables the placenta to keep working and keeps the uterine relaxed. It additionally helps

the mother's immune system tolerate the growth of the child as it is seen as foreign in the body.

- Relaxion → loosens the uterine muscles to prepare the body for birth.
- Oxytocin → Stimulates the production of milk and allows the uterine muscles to contract > allowing the body to go into labour and give birth.

| **Menstrual Cycle** | | |
|---|---|---|
| **Stage** | **Time Span (days)** | **Event** |
| **Menstruation** | 1-4 | Uterine (Uterus) bleeding, accompanied by shedding of the Endom |
| **Pre-ovulation** | 5-12 | Endometrial repair begins; development of ovarian follicle; uterine gradually thickens. |
| **Ovulation** | 13-15 | Rupture of mature follicle, releasing egg. |
| **Secretion** | 16-20 | Secretion of watery mucus by glands of the Endometrium, cervix, a uterine tubes; movement and breakdown of unfertilised egg; devel of corpus luteum. |
| **Pre-menstruation** | 21-28 | Deuteration of corpus luteum and Endometrium. |

# Oogenesis / Ovulation / Ovarian cycle

- Follicular phase
- Ovulation phase
- Luteal Phase